

- 1) a) checking all elements $\Rightarrow$ Ans (Linear search)
 b) rearranging the elements
 c) using hash values
 d) dividing array into halves

28
30

- 2) a) randomly order (Binary)
 b) sorted $\Rightarrow$ ans
 c) reverse
 d) empty

3) what is the middle index formula in BS (low to high)

a) $\frac{\text{low} + \text{high}}{2} \Rightarrow$ ans

b) $\frac{\text{low} - \text{high}}{2}$

c) $\frac{\text{high}}{2}$

d) $\frac{\text{low} + 1}{2}$

4) which data structure uses first in first out
 ans $\Rightarrow$ Queue.

5) last in first out
 ans $\Rightarrow$ Stack.

6) what is the primary function of stack a 'pop' operation

at ~~add an element~~ removes the top elements

7) In a linked list, each node contains:

- Data and link

8) The head pointer in a linked list points to

- X - last $\Rightarrow$ ans
- mid
- first
- null

9) The tail node in a singly linked list contains:

- Null in its next pointer $\Rightarrow$ ans
- pointer — previous node
- pointer — head node
- Random address

10) In a queue, insertion happens at:

- Front ~~is~~ ~~ans~~ X
- Rear ~~=~~ ~~ans~~
- middle
- Any position

11) In a queue, deletion happens at

- Rear $\Rightarrow$ ~~ans~~ X
- Front $\Rightarrow$ ans
- middle
- any position

12) which of the following is an application of stack?

- Job scheduling

- Infix to prefix convert $\Rightarrow$ ans
- process management
- Data buffering

13) which operation in linked list requires updating the previous node's pointer?

- Insert at begin
- Insert at mid
- Transversing the list access $\Rightarrow$ ans

14) Suffix array is used primarily for

- sorting integers
- String pattern match $\Rightarrow$ Ans:
- search in graphs
- Array compression ~~$\Rightarrow$ ans~~

15) CP array stores

- Diff between array ele
- length of CP bet consecutive suffix $\Rightarrow$ ans
- length of suffix
- length of repeated substrings

16) what type of data structure best repre

- stack

187) when inserting an element at specific position in an array, what must be done to preserve it

- Overwrite
- Shift subsequent $\rightarrow$ ans
- Resort the array
- reverse

188) If a binary

$\Rightarrow$ Ans: Target is not present in the sorted array.

191) while searching for 'cat' in a dictionary using binary search, what comparison logic should be applied between 'cat' and the middle word

- Compare alphabetical order

201) In a linked list, what happens if you forget to update the previous node's pointer during insertion?

- New node becomes unsearched node $\Rightarrow$ Ans

211) when reversing a singly linked list, which operation is essential at each step? - Delete each node

- Reverse the direction

221) when traversing a linked list to search for a value, how do you identify the end of the list?

- when next node is null.

23) to insert node at the beginning of a linked list.

which pointer must be updated first?

- Next of head node

24) which of the following correctly removes the front element of a queue implemented using linked list

- Set head pointer

25) A stack is used to check balanced parentheses in an expression. when do you pop from the stack?

- on reading a closing bracket.

26) when designing a postfix expression evaluator using a stack what should be pushed when a number is read?

- The operand value

27) while implementing queue operation, what condition signifies an empty queue?

- $Front == Rear$

28) for implementation undo functionality in a text editor, which data structure best suits this purpose?

- Stack.

29) which scenario is best suited for a queue structure?

Task scheduling by CPU

- Checking balanced expressions.

Q1 To remove duplicate characters from a string using an array. Which approach is appropriate?

- Compare each character with all previous and copy unique ones.